

Quiz 1 Solution

Course: CSC354 – Machine Learning
Semester: 7th (FA22-BCE-A/B)
Total Marks = 15

Date: 19-09-2025
Time Allowed: 30 min

Question 1: Relate the concepts of fundamentals of machine learning and linear regression to answer the following questions. (CLO-1, PLO-1, C1) (9 Marks)

- A. How many input variables are involved in the univariate linear regression? Write an expression for hypothesis of a univariate linear regression. (2)

Univariate linear regression involves one variable. The expression is $h_0(x) = \theta_0 + \theta_1 \cdot x$

- B. Predicting the atmospheric temperature at midnight every Sunday is a classification problem or a regression problem? (1)

This is a regression problem since atmospheric temperature is represented by a continuous value.

- C. Suppose you have a stock of 90 motorcycles manufactured by unknown companies. Your task is to arrange them into three groups based on the horse power feature. What kind of machine learning algorithm will you use? (1)

I shall use clustering (an unsupervised machine learning) algorithm, since the data needs to be converted into three groups based on the input feature of horse power.

- D. In the case of linear regression, how many minima points are expected in the cost function? (1)

The cost function of the linear regression algorithm has 1 minima point.

- E. While implementing linear regression with multiple variables, if the cost increases after every iteration, what should be done to correct it? (1)

The value of learning rate should be reduced.

- F. What is the benefit of converting all features on the same scale? (1)

The algorithm converges quickly.

- G. Is it possible to achieve a non-linear fit on the data using single feature? If No, why? If yes, how? (2)

Yes, this is possible. It can be achieved by adding polynomial features.

Question 2 (CLO-2, PLO-2, C3) (6 Marks)

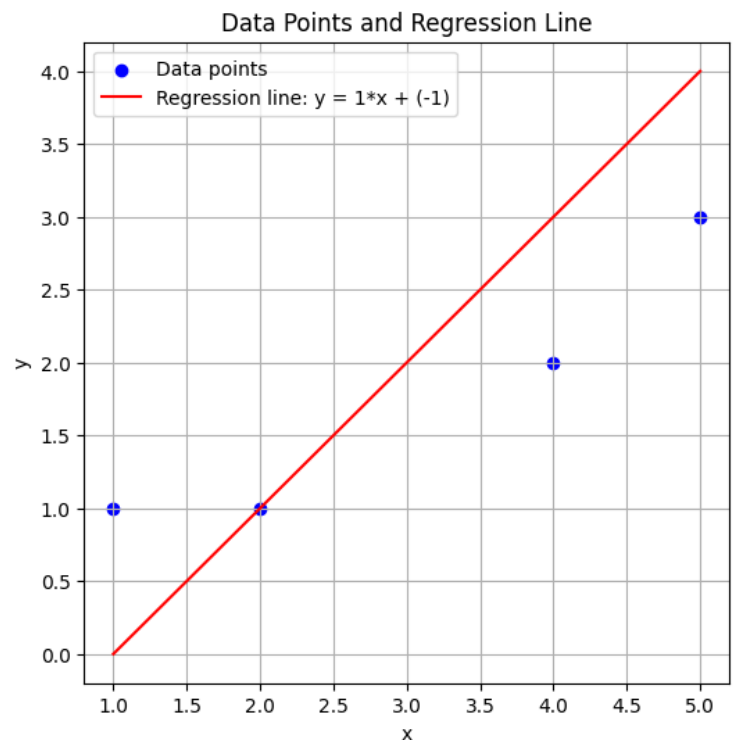
Given four examples (x, y) : (1,1), (2,1), (4,2), (5,3), and the parameters (θ_0, θ_1) : (-1,1), compute the predicted output for each example and the overall cost (mean squared error). Also Plot the data and show the regression line with labelled coordinate axes. (6)

Data:

$(x_1, y_1) = (1,1)$,
 $(x_2, y_2) = (2,1)$,
 $(x_3, y_3) = (4,2)$,
 $(x_4, y_4) = (5,3)$

Parameters:

$\theta_0 = -1$



$$\theta_1 = 1$$

Predictions:

$$\hat{y}^i = \theta_0 + \theta_1 \cdot x^i$$

So:

$$\hat{Y}^1 = -1 + 1 \cdot 1 = 0$$

$$\hat{Y}^2 = -1 + 1 \cdot 2 = 1$$

$$\hat{Y}^3 = -1 + 1 \cdot 4 = 3$$

$$\hat{Y}^4 = -1 + 1 \cdot 5 = 4$$

Squared Errors:

$$(e^1)^2 = (y^1 - \hat{y}^1)^2 = (1 - 0)^2 = 1$$

$$(e^2)^2 = (y^2 - \hat{y}^2)^2 = (1 - 1)^2 = 0$$

$$(e^3)^2 = (y^3 - \hat{y}^3)^2 = (2 - 3)^2 = 1$$

$$(e^4)^2 = (y^4 - \hat{y}^4)^2 = (3 - 4)^2 = 1$$

Mean Squared Error:

$$\text{MSE} = (1/4) \cdot [1 + 0 + 1 + 1] = 3/4 = 0.75$$